

CHISHOLM-HIBBING, MN, USA

WMO: 727455

Lat: 47.386N

Lon: 92.839W

Elev: 412

StdP: 96.47

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.9	-34.0	0.2	-30.7	-31.2	0.2	-27.4	11.1	-13.0	10.0	-13.0	1.5	320	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	29.6	21.0	27.9	19.8	26.5	18.8	22.7	27.8	21.4	26.1	20.2	24.6	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.4	25.8	19.7	15.2	24.3	18.5	14.1	22.8	69.0	27.9	63.9	25.9	59.7	24.5	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.5	-35.3	32.2	3.6	1.6	-37.9	33.3	-40.0	34.2	-42.0	35.1	-44.6	36.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-34.9	24.7	3.1	1.0	-37.2	25.4	-39.0	26.0	-40.7	26.5	-43.0	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-13.6	-11.2	-3.9	3.7	10.5	15.8	18.4	17.2	12.8	5.7	-2.5	-10.1
	DBStd	12.45	7.85	7.39	7.08	5.10	4.52	3.72	3.17	3.29	4.59	4.60	5.97	7.13
	HDD10.0	3175	731	593	432	197	48	2	0	0	23	149	376	624
	HDD18.3	5457	989	826	689	439	248	93	38	61	174	392	626	882
	CDD10.0	856	0	0	1	7	62	177	262	224	107	16	1	0
	CDD18.3	96	0	0	0	0	3	17	42	26	8	0	0	0
	CDH23.3	1339	0	0	0	4	93	261	538	335	101	7	0	0
	CDH26.7	296	0	0	0	0	18	59	131	69	18	1	0	0
Wind	WSAvg	3.2	3.3	3.4	3.5	3.7	3.5	3.0	2.7	2.6	2.8	3.3	3.3	3.2
Precipitation	PrecAvg	646	17	12	24	41	70	100	102	87	77	58	31	19
	PrecMax	828	59	41	59	111	141	167	243	183	162	144	91	38
	PrecMin	444	0	0	2	6	10	20	25	14	13	11	2	1
	PrecStd	102	13	10	14	26	33	40	52	46	37	39	23	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	7.4	17.5	23.0	29.0	31.2	31.8	31.2	28.8	24.1	15.4	5.5
		MCWB	1.7	3.9	10.6	13.1	18.2	22.6	23.0	22.6	20.4	15.5	10.4	2.5
	2%	DB	1.3	4.2	12.1	19.4	26.1	28.1	29.8	28.4	26.3	19.9	11.7	2.8
		MCWB	-0.1	2.0	7.0	10.3	15.7	19.6	21.2	20.6	19.0	13.3	8.0	1.2
	5%	DB	-0.4	2.2	8.7	16.7	23.0	26.3	28.0	26.9	23.9	17.0	8.6	1.2
		MCWB	-1.3	0.4	4.6	8.5	14.3	18.4	20.4	19.4	17.5	12.4	5.4	0.1
	10%	DB	-2.4	-0.1	5.8	13.7	20.4	24.3	26.5	25.1	21.7	13.9	6.1	-0.2
		MCWB	-3.4	-1.4	2.8	7.1	12.7	17.1	19.3	18.3	16.5	10.1	3.9	-1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	4.4	12.6	14.5	20.1	23.6	24.4	23.8	21.9	17.3	11.8	3.3
		MCDB	3.9	6.5	15.5	20.7	26.0	29.1	29.6	29.0	26.8	21.3	13.7	4.3
	2%	WB	0.2	2.5	7.6	11.2	17.5	21.3	23.0	22.0	20.2	14.9	8.3	1.4
		MCDB	1.3	4.2	10.7	17.9	23.5	26.8	28.3	26.8	24.3	18.9	11.0	2.6
	5%	WB	-1.5	0.4	5.2	9.2	15.6	19.7	21.6	20.6	18.8	12.7	5.8	0.2
		MCDB	-0.5	1.9	8.4	15.5	21.1	24.0	26.2	25.0	22.6	16.2	8.3	1.1
	10%	WB	-3.3	-1.6	2.9	7.5	13.8	18.2	20.3	19.5	17.4	10.5	3.8	-1.1
		MCDB	-2.3	-0.1	5.7	13.0	19.1	22.6	24.7	23.7	20.7	13.5	5.8	-0.3
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.1	12.0	12.8	14.4	13.7	13.8	13.7	13.2	10.8	9.0	9.3
		MCDBR	10.5	12.8	14.8	19.4	19.5	16.4	15.7	15.8	16.5	16.8	13.7	10.4
		MCWBR	9.4	10.2	10.1	11.6	10.6	9.0	8.4	8.6	9.8	10.7	9.8	8.9
	5% WB	MCDBR	10.3	11.5	14.0	17.4	16.2	13.6	13.8	13.6	13.2	14.7	12.4	9.6
		MCWBR	9.3	9.4	9.7	11.0	9.6	8.4	8.2	8.3	8.4	10.2	9.4	8.4
Clear-Sky Solar Irradiance	taub		0.247	0.260	0.308	0.347	0.364	0.371	0.387	0.376	0.352	0.308	0.283	0.261
	taud		2.372	2.340	2.358	2.372	2.383	2.385	2.357	2.417	2.458	2.572	2.521	2.401
	Ebn at Noon		858	919	916	907	900	892	871	867	858	857	811	792
	Edh at Noon		68	88	103	112	116	116	118	107	93	71	60	60
All-Sky Solar Radiation	RadAvg		1.43	2.57	3.72	4.64	5.25	5.62	6.04	5.10	3.80	2.27	1.32	1.01
	RadStd		0.17	0.24	0.35	0.47	0.38	0.42	0.27	0.39	0.29	0.31	0.17	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page